

HK DSE Biology â Photosynthesis (detailed test notes)

1. Overview

Photosynthesis converts light energy into chemical energy in green plants, algae, and some bacteria. Carbon dioxide and water become glucose and oxygen.
Word equation: carbon dioxide + water $\rightarrow$ glucose + oxygen (light, chlorophyll).

2. Where it happens

Leaf palisade cells are packed with chloroplasts. Chlorophyll a and b in the thylakoid membranes absorb red and blue light and reflect green.
Stomata on the lower epidermis let CO₂ in and O₂ and water vapour out.
Xylem brings water from roots. Phloem exports sugars as sucrose.

3. Light-dependent stage (thylakoid / grana)

Photolysis splits water: $2\text{H}_2\text{O} \rightarrow 4\text{H}^+ + 4\text{e}^- + \text{O}_2$. Electrons enter photosystem II, then photosystem I. NADP⁺ is reduced to NADPH. ATP is made by chemiosmosis through ATP synthase (photophosphorylation). Oxygen diffuses out via stomata.

4. Light-independent stage (Calvin cycle, stroma)

Rubisco fixes CO₂ onto ribulose biphosphate (RuBP). The product is reduced using ATP and NADPH to triose phosphate. Some triose becomes glucose; the rest regenerates RuBP. Temperature, CO₂ concentration, and light intensity can each limit the rate (limiting factors).

5. Limiting factors (exam points)

- Low light: fewer electrons and less ATP/NADPH, so Calvin cycle slows.
- Low CO₂: rubisco has less substrate; extra light cannot raise the rate.
- Low temperature: enzyme (rubisco) collisions slow; rate falls.

A graph of rate vs light rises then plateaus when CO₂ or temperature limits.

6. Adaptations of the leaf

Broad lamina captures light. Thin blade shortens the diffusion path.

Waxy cuticle cuts water loss. Air spaces in the spongy mesophyll hold CO₂.

Guard cells open stomata in light when photosynthesis uses CO₂.

7. Compare respiration

Photosynthesis stores energy; aerobic respiration in mitochondria releases it. In light, net gas exchange is O₂ out and CO₂ in. In the dark, only respiration. Compensation point: photosynthesis rate equals respiration rate.

8. Common DSE mistakes

Chlorophyll does not 'make' food by itself; it captures light for the reactions.

Oxygen comes from water photolysis, not from carbon dioxide.

Glucose is often converted to starch for storage or sucrose for transport.

9. Quick check

Name the two stages, the organelle, the pigment, the gas in, the gas out, and one limiting factor. Explain why a brightly lit sealed tube of pondweed produces bubbles faster when sodium hydrogencarbonate is added (more CO₂).